

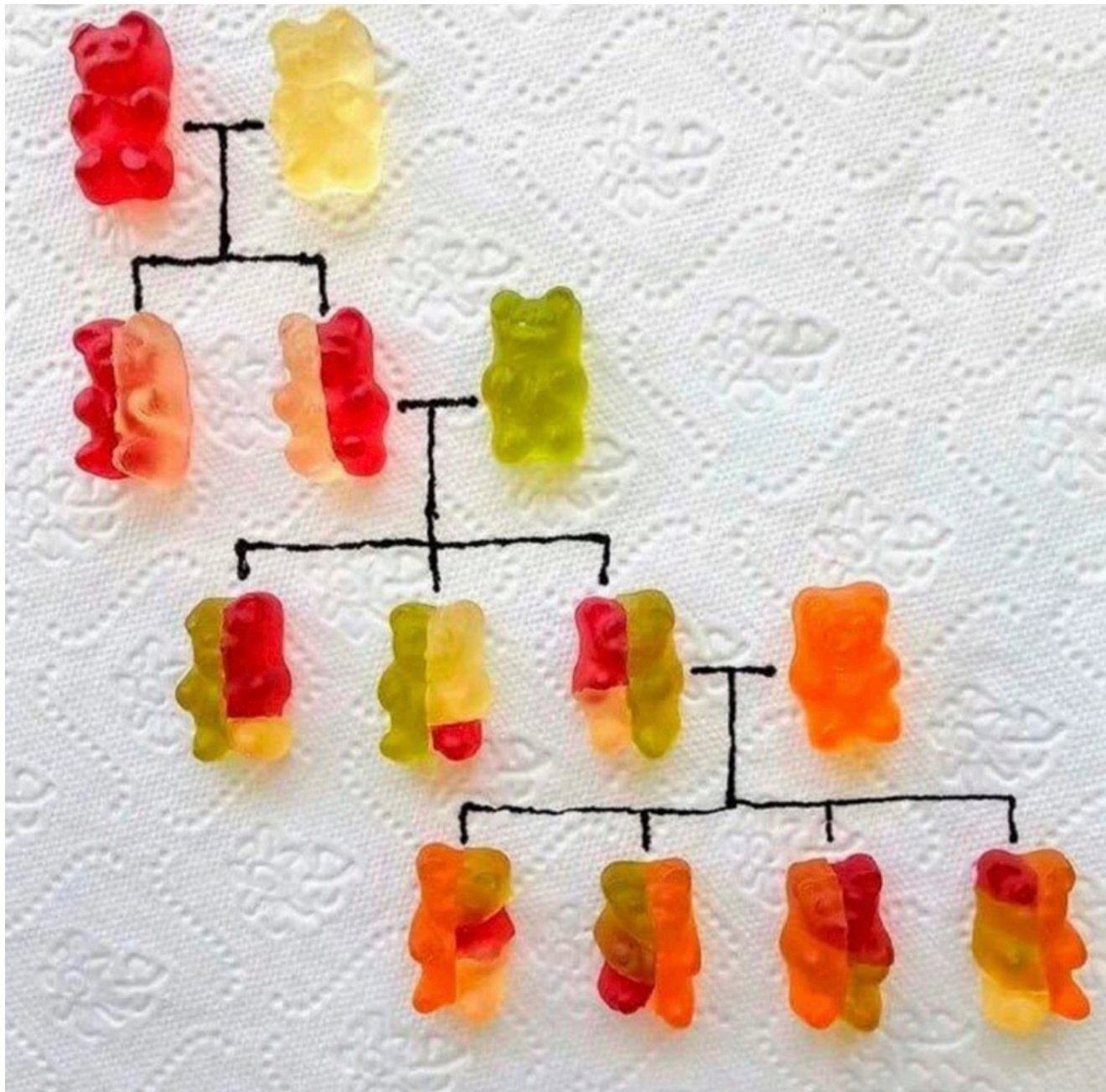


**What's going on from
-9 to +3 months?**

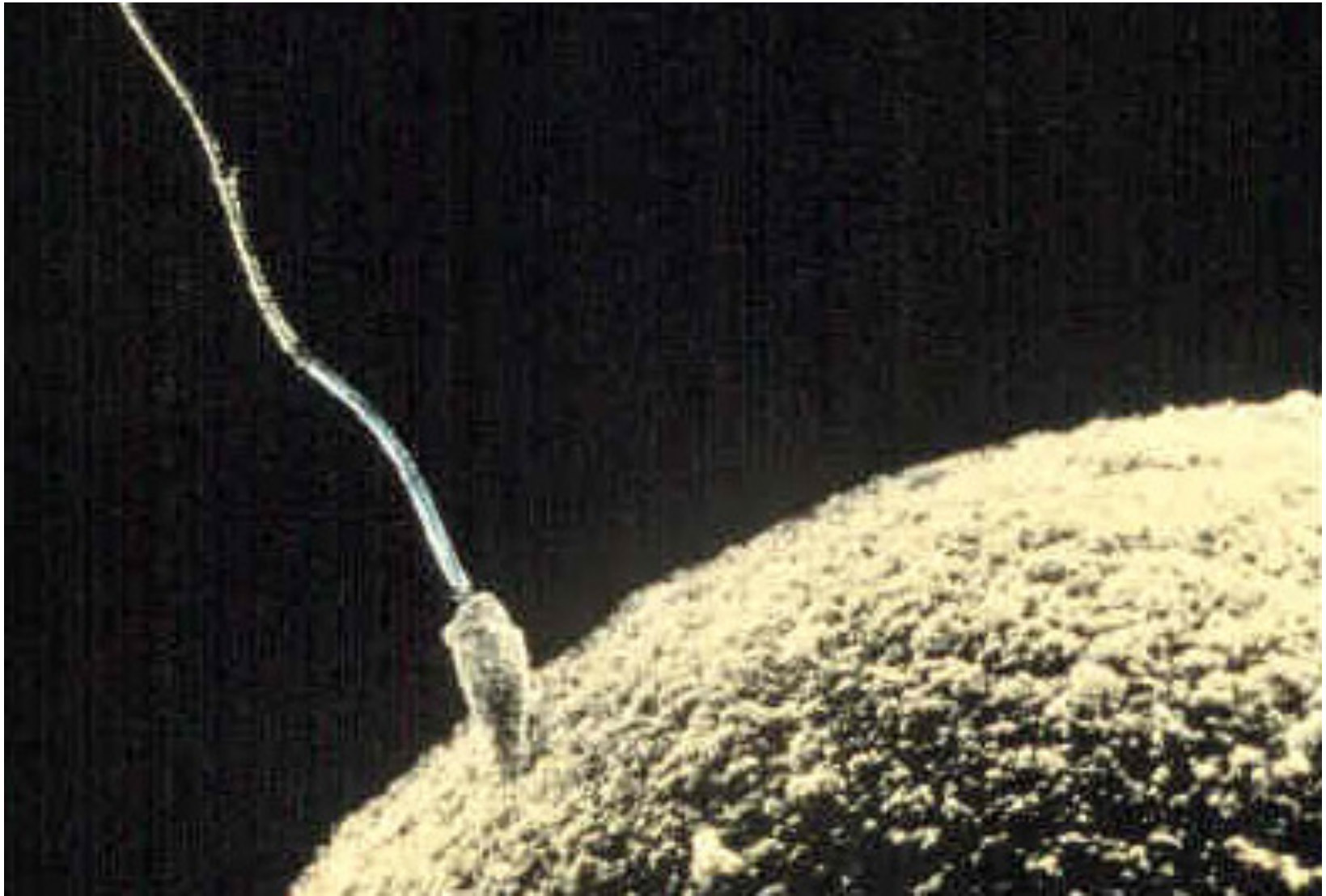
T/Th 9:35-10:25am

Professor Casey Lew-Williams

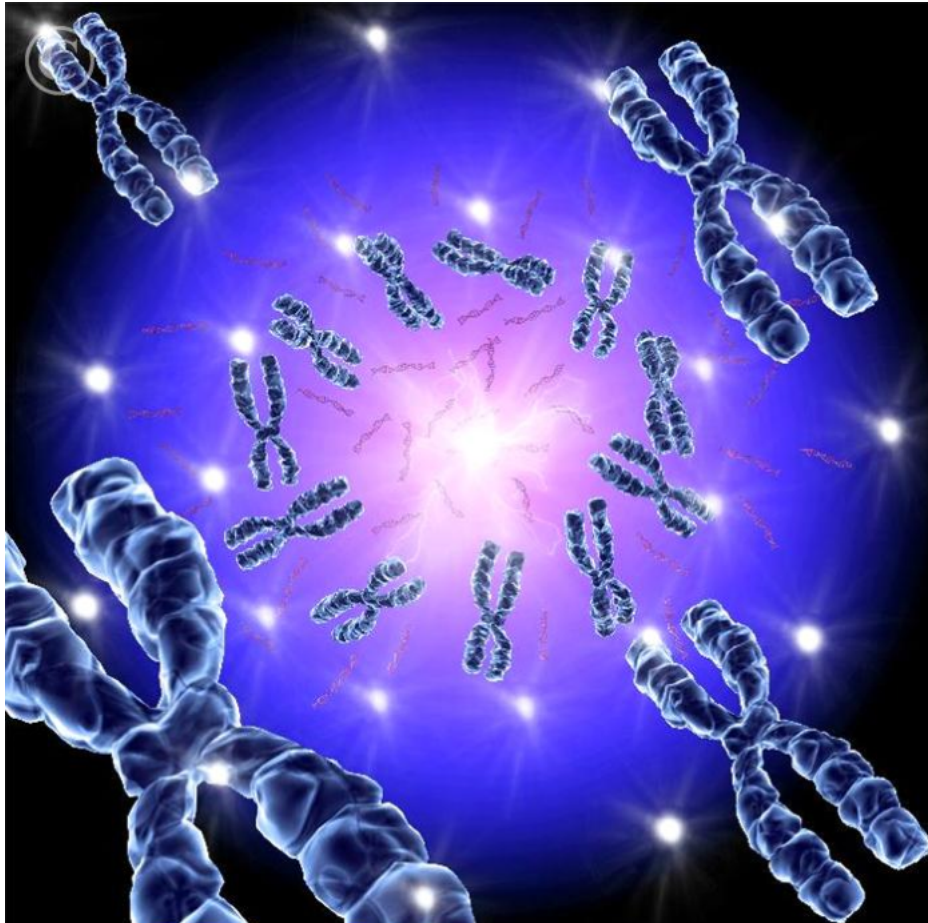
Kennedy Casey, Abby Fergus, Sarah Joo, Chantal Valdivia, Claire Whiting

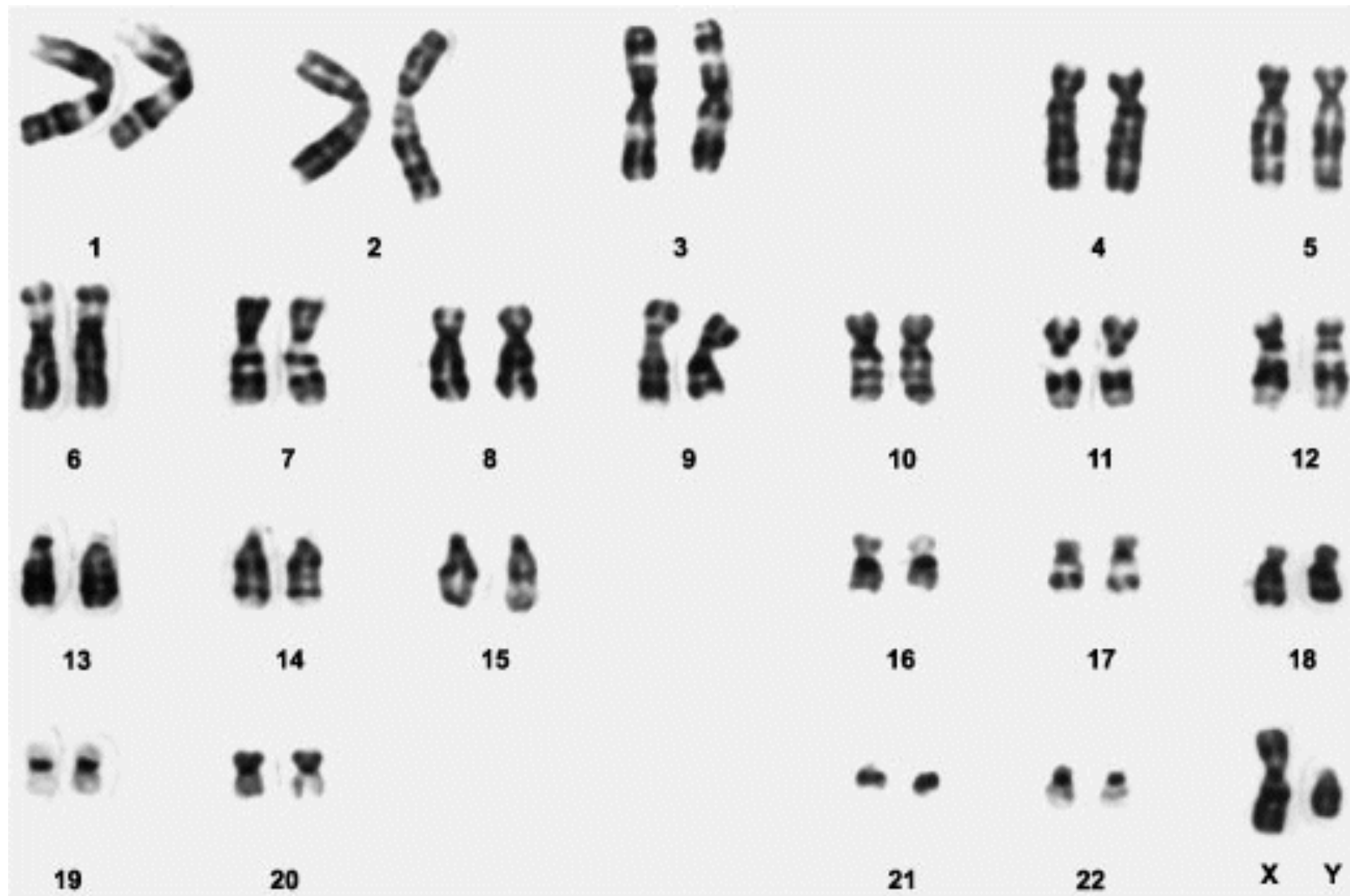


When does development begin?









17,600,000,000,000 combinations

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309,000,000,000,000,000,000,000,000,000



Embryo at 4 weeks



Face development from 5.5-8.5 weeks



9 weeks marks the transition from embryo to fetus



NEWS IN BRIEF

**Michelle Obama 'Becoming' Netflix
Documentary Spends First Hour On
Embryonic Stage**

Fetus at 11 weeks



Fetus at 16 weeks

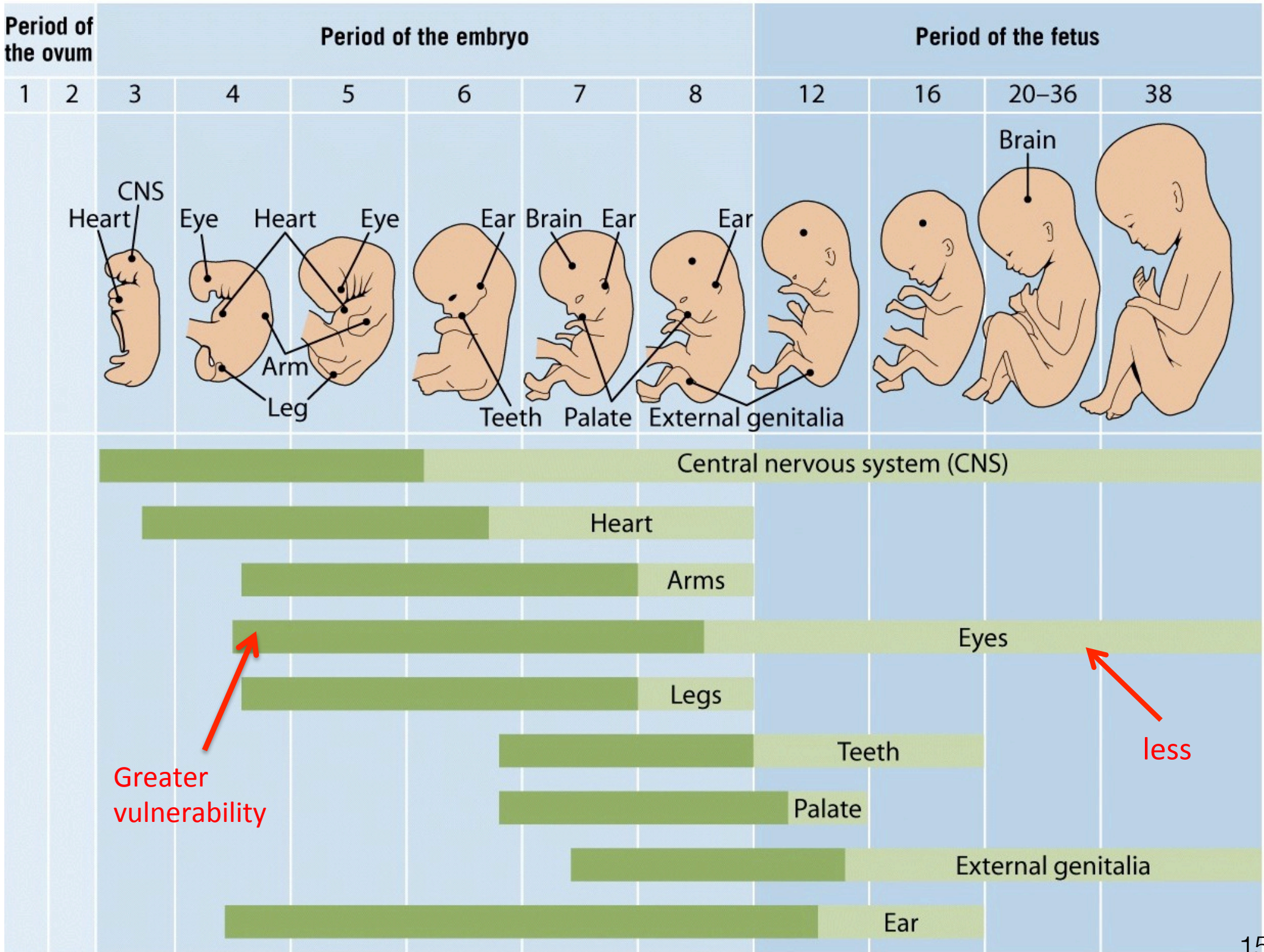


Fetus at 18 weeks



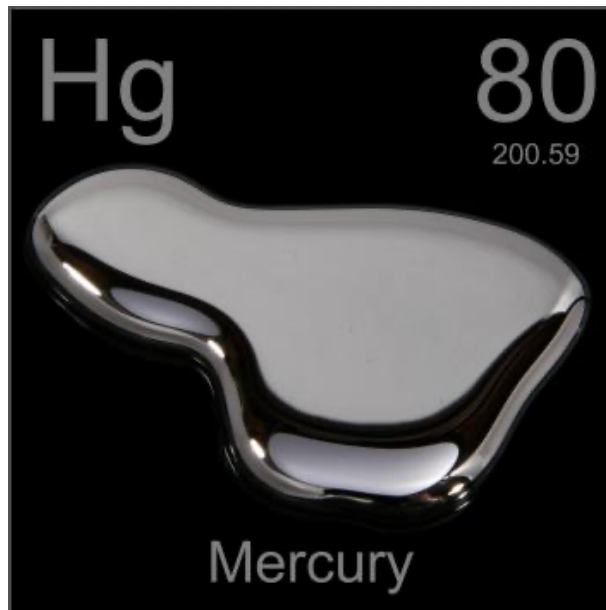
Fetus at 28 weeks





Environmental effects on prenatal development

Teratogens



Fetal alcohol spectrum disorders (FASD)

- More exposure = more effects

- Minor facial abnormalities
- Slow growth
- Cognitive deficits
- ADHD



Mercury



第10回患者・岩坂さん子10歳。本病院のため兄は2歳、姉は6歳で死亡。彼女は30歳となったが、その生涯の大半を施設で暮らしている。

Photo by Takeru Shima 1971

10歳



Long-term effects of prenatal environment

- Excessive maternal weight gain in pregnancy associated with obesity, diabetes, possibly some cancers
- Maternal anxiety, depression, stress associated with behavior challenges in child 4 years later
- Maternal stress is also weakly linked to mental health difficulties (e.g., schizophrenia)



A very interesting Radio Lab episode:

Inheritance

<http://www.radiolab.org/2012/nov/19/>



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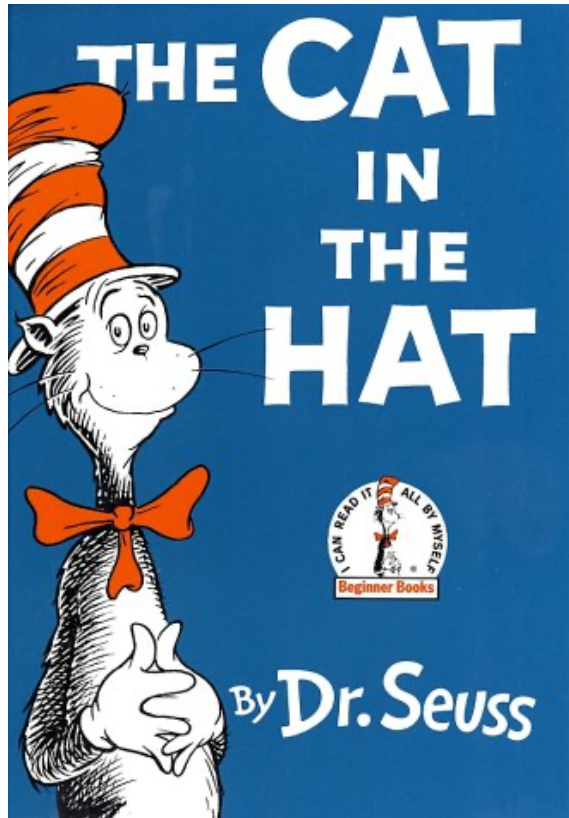
Sold by [Scharf Industries LLC](#)
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Add a Protection Plan:

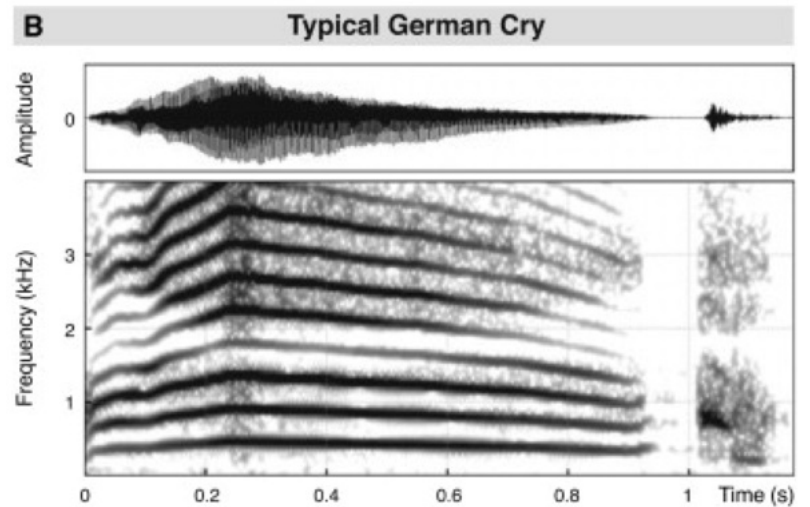
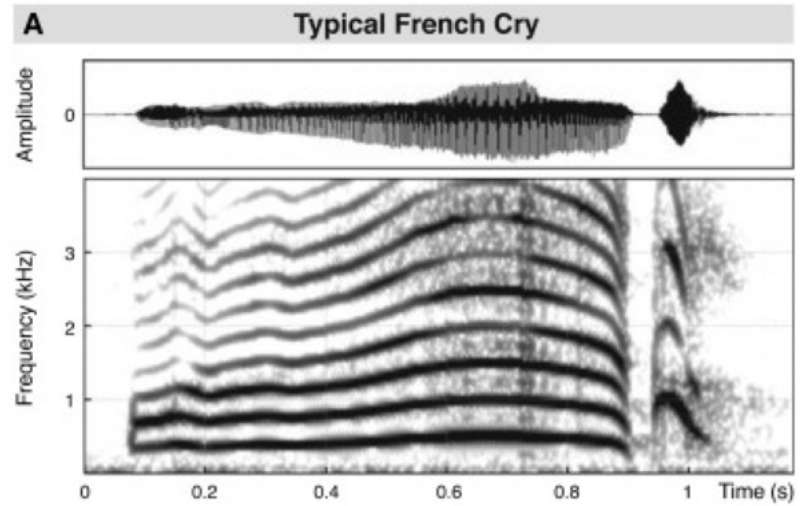
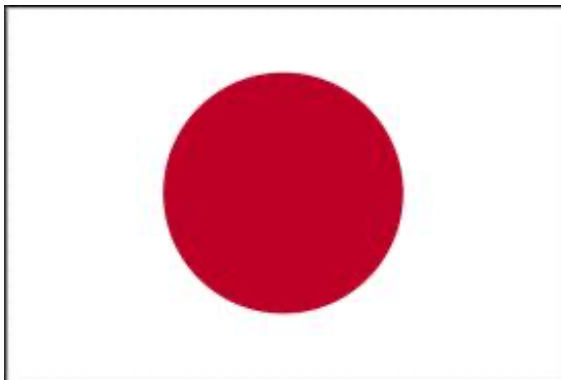
- ☐ **3-Year Protection** for **\$21.99**
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Roll over image to zoom in





I sat there with Sally, we sat there we two.
And I said, "How I wish we had something to do!"
Too wet to go out and too cold to play ball.
So we sat in the house. We did nothing at all.
So all we could do was to Sit! Sit! Sit! Sit!
And we did not like it. Not one little bit.
And then something went BUMP!
How that bump made us jump!
We looked! Then we saw him step in on the mat!
We looked! And we saw him! The Cat in the Hat!



Prenatal music exposure
facilitates neural tuning
to key dimensions of
human speech

(Arenillas-Alcón et al., 2023)





Newborn Soothed By Familiar Sound Of Parents' Bickering

NEWS IN BRIEF • Family • Local • Kids • ISSUE 50•18 • May 5, 2014





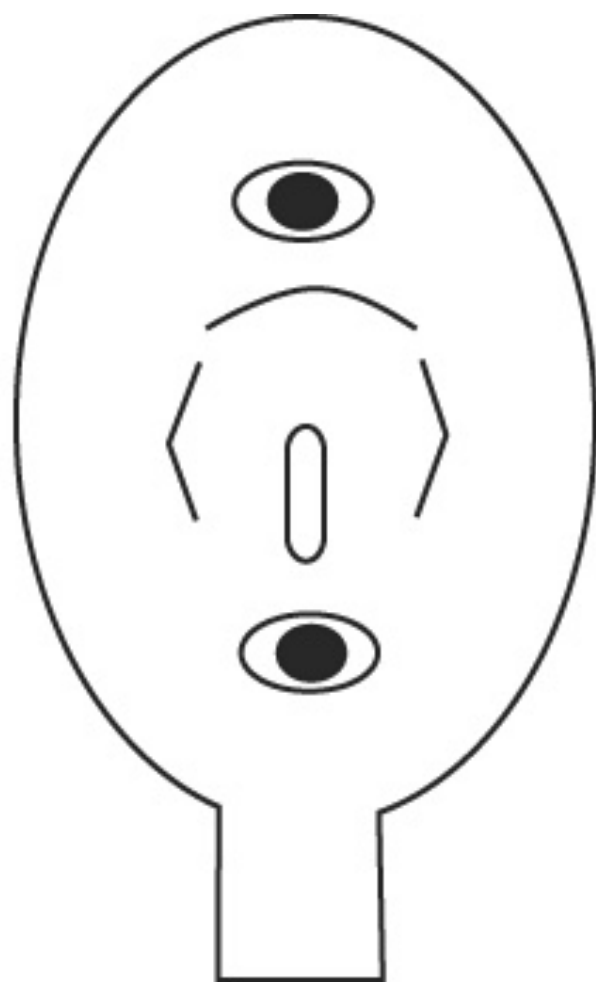
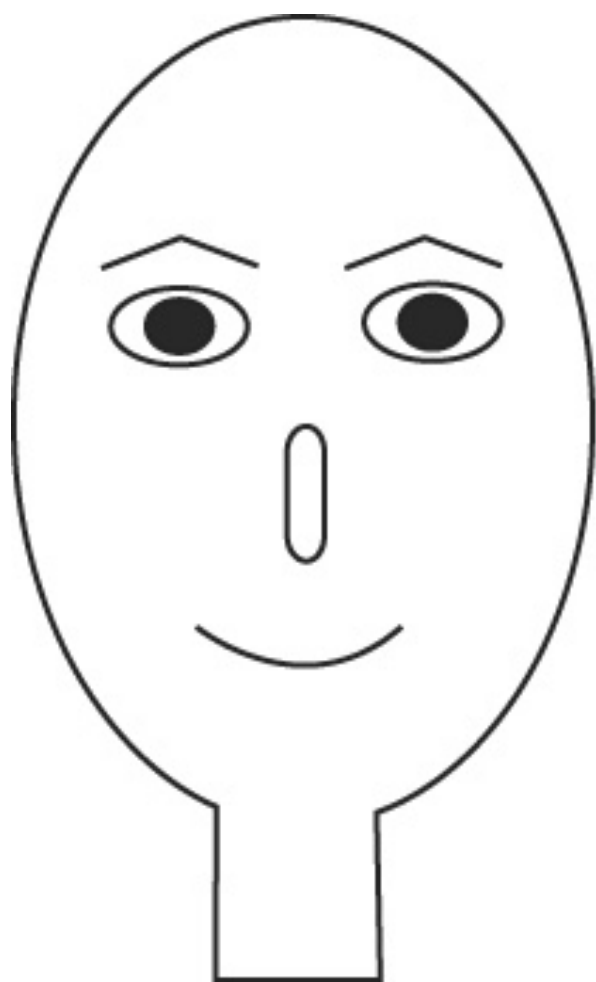
NEWS IN BRIEF

Study Finds Newborn Infants Can Tell If Parents Are Losers

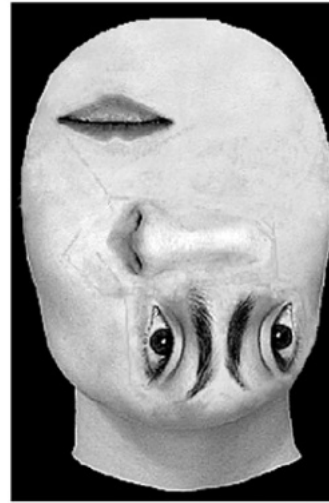
3/21/12 2:15pm • SEE MORE: SCIENCE & TECHNOLOGY ▾

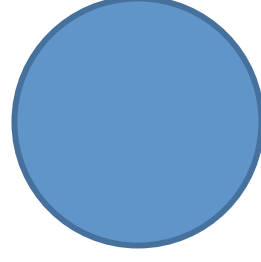
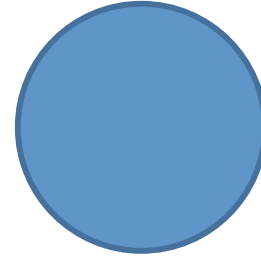
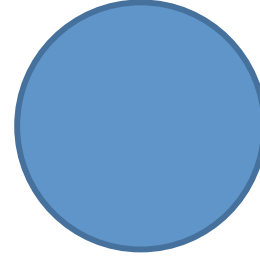
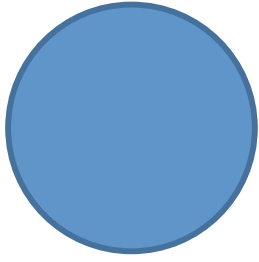
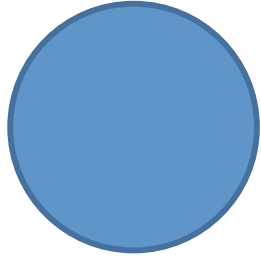
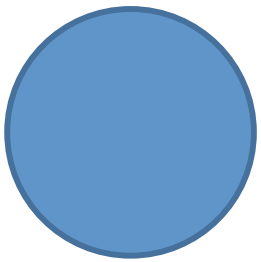


CHICAGO—A study published this week in the journal *Pediatrics* found that, within seconds of their birth, babies have the ability to sense whether their parents are losers. "From the moment they open their eyes, newborns can tell if their mother had no other options and was forced to settle for their









The Nativist - Empiricist Debate



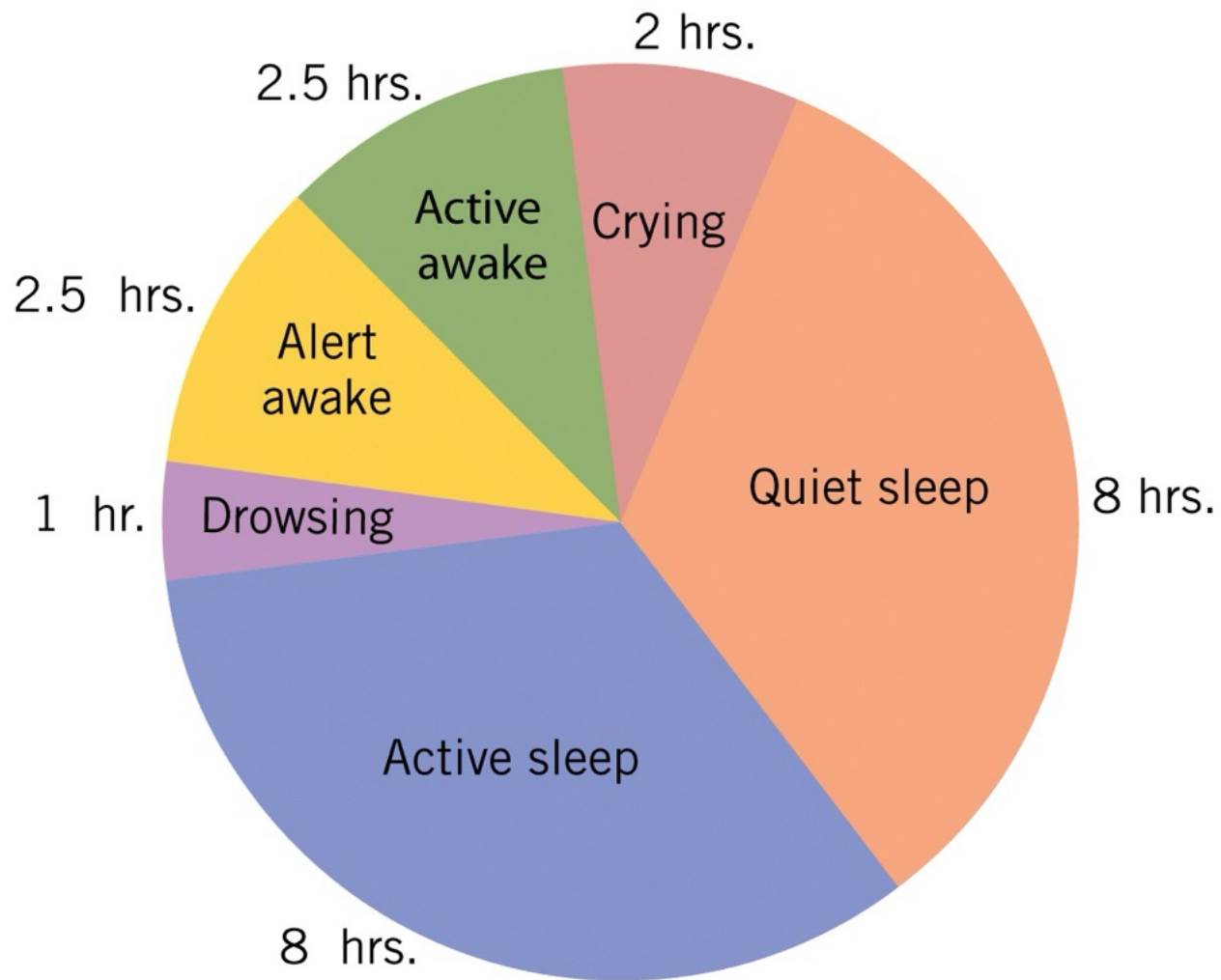
Timeline of the first hour of life

- **Minute 0:** cry
- **Minute 2:** relax
- **Minute 2.5:** open eyes, move head, move mouth
- **Minute 8:** interested in eating, look at mother's face and breasts, move hands toward mouth
- **Minute 18:** relax
- **Minute 36:** re-energized, scoot toward breasts (mostly using smell to navigate)
- **Minute 62:** nursing, ingestion of colostrum
- **Minute 70:** sleep

Newborn states



The newborn



Newborn states

Drowsiness
Active sleep
Quiet sleep
Alert awake
Active awake
Crying

